



DEALER BEST PRACTICE SERIES

Temporary Rim & Tires for Shop Transportation and Support

Application Maintenance Site Management Component Rebuild Component Life Management Safety MARC Management

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1.0 Introduction

Assembly and rebuild procedures for Off Highway Trucks present a number of challenges due to their size, weight and the equipment required to carry out these procedures. This best practice introduces a custom rim and tire group intended to facilitate the rebuild process.

During the assembly or rebuilding process, Off Highway Trucks need to be adequately supported and may need to be transported to various locations within the dealership to accommodate the assembly progression. Employing this rim and tire combination facilitates and improves build efficiency and improves machine transportation when compared to traditional methods.

2.0 Best Practice Description

These custom rims were developed by WesTrac, Guildford branch, to support the execution of LOHT rebuilds for mine sites located within their territory.

Dealer assembly/rebuild procedures require machines to be adequately supported using a safe, efficient and effective method. At present stands are the most commonly employed supporting device; these stands provide adequate support but cannot be used during machine transportation. A common procedure for the transportation and relocation of a machine requires the use of cranes for partially assembled machines and a set of standard rims & tires for an assembled machine. This custom rim and tire group serves two important purposes: they can be used as a supporting tool during machine assembly/rebuild (See Figures 1 & 2) and can be used for transportation purposes. As an example assembled Off Highway Trucks can be driven around the dealer yard on the custom rims and tires, eliminating the need for the standard rim and tire groups.



Figure 1 – Custom Rim and Tire Package

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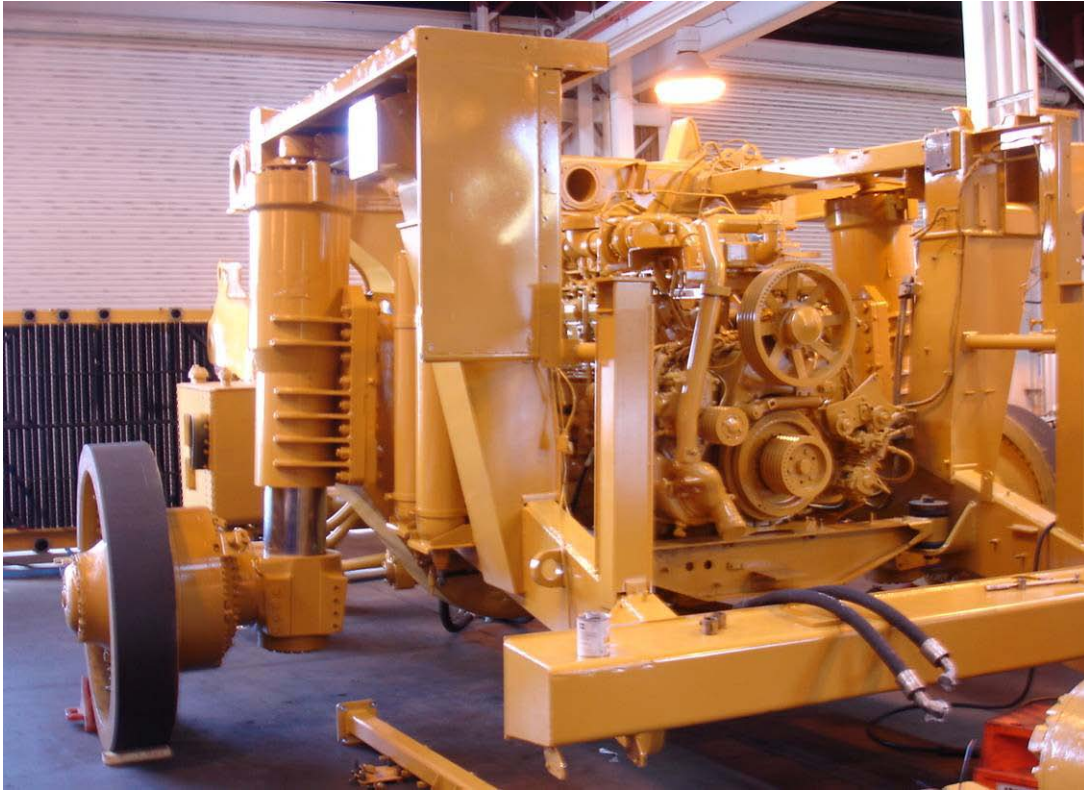


Figure 2 – Custom Rim and Tire Group employed during machine rebuilds

3.0 Implementation Steps

- Fabricate custom rim group based on machine model (See attachments for stud patterns).
- Contact local supplier to manufacture and install custom rubber tires (solid) onto custom rims.
- Introduce a modified assembly/rebuild procedure to incorporate this custom rim and tire package.

4.0 Benefits

- Safety - provides excellent machine support during the assembly/rebuild process and reduces working height for technicians.
- Resources & Efficiency - custom rim and tire group is significantly easier to remove and install when compared to the conventional rim and tire package. Special equipment such as a tire handler is not required for this procedure.
- Resources & Efficiency - custom rim and tire group provides easy access to the machine and eliminates the need for stands.
- Resources & Efficiency - machines can be easily transported and driven on custom rim and tire groups without the need for additional equipment, such as overhead cranes.
- Storage - custom rim and tire group is compact, little storage space is required when compared to the standard rim and tire combination.

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- Cost - global tire shortage makes it difficult for a dealer to have a set of dedicated tires for assembly/rebuild purposes within the Dealership
- Cost - one set of standard rim and tire groups for the assembly yard may be insufficient based on the number of machines being assembled or rebuilt. (See Figure 3)

5.0 Resources Required

- Weld/fabrication shop to manufacture custom rims.
- Capital for material and labor. Cost of urethane coated rims (one set) is estimated at AUD \$65,000.
- Supplier for custom solid rubber tires.



Figure 3 – Machine Assembly Area

6.0 Supporting Attachments / References

- It is recommended that these rims be manufactured based on JIS standards:
 - o JIS G3101
 - o JIS G3106
 - o JIS G4051
- Urethane coated rims supplier:
Downer EDI Mining – Rimtec P/L
 13 Fairbrother Street
 Belmont, Perth
 Western Australia 6104
 +61 08 94781333
 Email: rimtec@downeredimining.com
 Website: www.rimtec.com.au

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7.0 Related Best Practices

None

8.0 Acknowledgements

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